IDENTIFYING A DRILLING STATE MACHINE

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Contents

Abbreviations

AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
GPM	Gallons per Minute
HMM	Hidden Markov Model
MFI	Mud Flow Index
MR	Motor (on a drill string)
MWD	Measurement While Drilling
PCA	Primary Component Analysis
PC1	Primary Component 1
PC2	Primary Component 2
PD	Pulser Device
ROP	Rate of Penetration
RPM	Revolutions per Minute
RT	Real Time
WOB	Weight on Bit

Executive Summary

The client, Deere Development Company, has requested a method for predicting the drilling state machine (as defined by them) based on typical downhole drilling parameters for flow, rotation, vibration, and WOB for oil rig drilling. The drilling state machine defines five states: Pumps Off, Pumps On, Sliding Drilling, Rotating, and Rotational Drilling. By identifying the state correctly, the client can better implement other software for drilling optimization. This combination will make oil rig work more cost effective, time efficient, and safe for workers. An optimally fitted Hidden Markov Model was defined by seven states and using only WOB time-series data. Unfortunately, this model was not interpretable and therefore cannot be used in real-world applications. Therefore, future work on this project should consider Gaussian Mixture Model (GMM) clustering, or gathering accurately labeled data and skipping straight to designing a Random Forest (RF) that may be implemented directly in the client's software.

Introduction

This project focuses on using drilling parameters of oil rig drill strings to increase the knowledge and efficiency of runs. The client, Deere Development Company, has requested a method of estimating the drilling state machine, as defined by them, based on the drilling parameters sensed by the tool in real time. The knowledge of the drilling state machine can then be utilized to turn on and off other software that optimizes drilling efficiency and safety, critically in terms of energy, cost, and time.

The drilling state machine has been defined with five modes. High and Low are relative terms to describe how each parameter changes from the last drilling state. The drilling state machine can only move consecutively with the exception of (1) providing access to (2) and (3), but (2) not providing access to (3) (see Figure 1) (Deere, 2025).

Drilling Parameter	Flow	Rotation	Vibration	WOB
(0) Pumps Off	Low	Low	Low	Low
(1) Pumps On	High	Low	High	Low
(2) Sliding Drilling	High	Low	High	High
(3) Rotating	High	High	High	Low
(4) Rotational Drilling	High	High	High	High

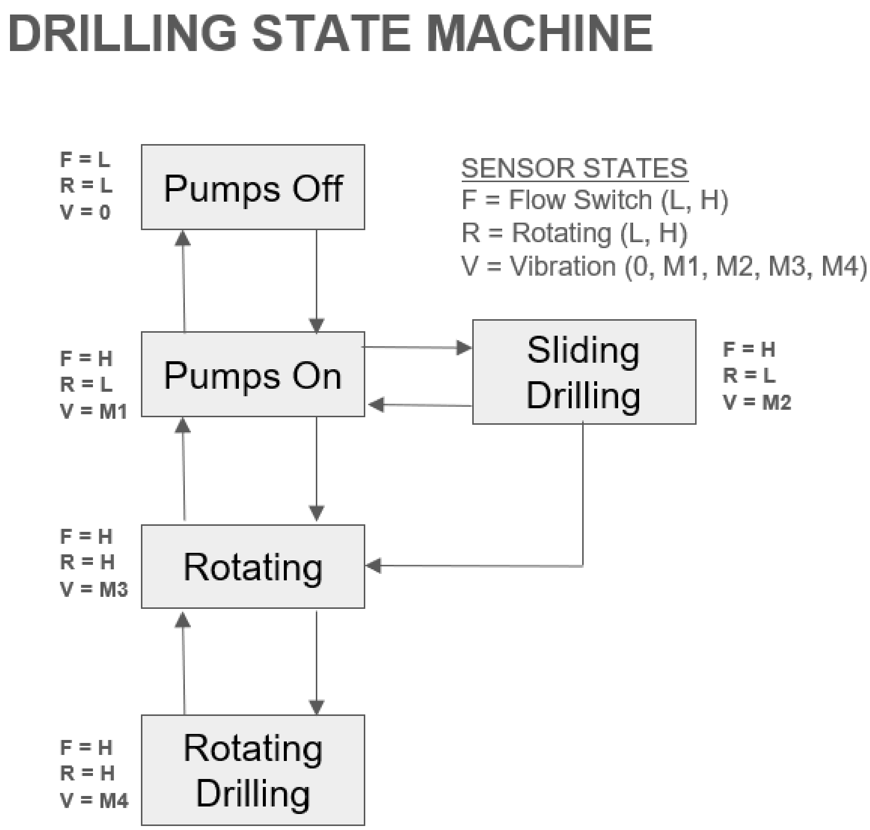

Figure 1 Deere Development Company's Drilling State Machine

This project initially intended to focus on using validation data to create labels and then use supervised learning methods, such as decision trees and support vector machines, to determine the drilling state machine at any given time during a run. However, after preprocessing the data, the validator columns showed signs of insufficient data for use as accurate labeling. Therefore, this project was reformed as an unsupervised approach to labeling drilling parameter data as parts of a drilling state machine.

The primary goal was to test variations of a sequential modeling method for adding the drilling state parameter labels to the training data. Hidden Markov Model (HMM) was used to address the time-series nature of the data and potentially provide more accurate labels for the training data (Jurafsky and Martin, 2024). This is particularly well suited for this project as we already have a Markov chain, namely the prepared drilling state machine.

Related Work

The problem of using drilling parameters to predict things about the downhole environment has become popular recently, especially for predicting geological formations. These projects have similar parameters to those proposed in this project to solve other problems. For instance, Hassan et al. (2024) tested three different models using ROP, GPM (flow), RPM, strokes per minute, torque, and WOB to predict the porosity and permeability of the geological formations being drilled. Of decision tree (DT), random forest (RF), and support vector machine (SVM) models, each had a correlation coefficient of above 0.8, but RF performed the best with 0.92. This is likely due to the robustness built into RF, since it uses the aggregate of multiple decision trees. A visualization on the prior work done on predicting geological formations from drilling parameters can be found in Hassan et al. (2024, p. 17068). These methods range from supervised to unsupervised learning.

Unsupervised learning techniques are also relevant for a range of research topics, both within the oil and gas industry and outside of it. For instance, Tashnizi (2022) wrote his master's thesis on finding outliers in pump jack performance to identify failures and optimize future performance. During this research, he tested K-Means clustering, Gaussian Mixture Model (GMM), and Hidden Markov Model (HMM) with a GMM implementation. Both GMM and HMM outperformed K-Means in accuracy for cluster sets larger than four (Tashnizi, 2022, p. 67).

This project will build on the findings of these experiments by focusing on unsupervised learning techniques for labeling data and applying the drilling parameters to predict the drilling state machine rather than the surrounding environment or torque. This technology will be more applicable to a wider range of associated technologies for optimizing drilling in terms of efficiency, safety, and accuracy.

Methodology

Datasets

The datasets for this project have been provided by the client, Deere Development Company. They represent six runs from three different oil rigs, named Flybar 1WB, Flybar 1WC, and Flybar 2WC. These datasets include time and depth as well as the following attributes relevant to this project. Each item indicated below represents a column in the dataset that will be used for the project.

DRILLING PARAMETER	ORIGINAL DATA	PREPROCESSED DATA	PROCESSING CHANGES
TIME	RT_Time	Time	Rename, Fill Down
DEPTH	RT_Depth	Depth	Rename, Fill Down
FLOW	• RT_Pumps_Off • RT_Pumps_On	GPM	Aggregate, Fill Down
ROTATION	MR_RPM-AVG.MR	Motor_RPM	Rename, Fill Down
VIBRATION	• MR_VIBA, MR_VIBL • PD_Axial Vibration, PD_Lateral Vibration	• Motor_Axial_Vibration • Motor_Lateral_Vibration • Pulser_Axial_Vibration • Pulser_Lateral_Vibration	Rename, Fill Down
WOB	RT_WOB	WOB	Rename, Fill Down

Preprocessing

The data was cleaned such that it includes the parameters seen in the previous table with no missing values. To avoid null values, the last value was repeated until a new one was found. If no value was found in the first row, a 0 was inserted. After the data was cleaned, it was segmented into each run specified by well.

Warehousing

As a final step in the preprocessing code, the data frames for each rig (in its entirety) and run were saved to CSVs in a private folder to protect the information of both the client and their customers.

Modeling

The Gaussian HMM module from the hmmlearn package, which is supported by scikit-learn, was used to implement the models. The Gaussian HMM was most appropriate due to the states' single modal nature and continuous data. A total of 700 models were created, tested, and evaluated for their log likelihood, AIC, and BIC. The models were created by combining the following parameters for each possibility:

- Random Seed: [792022, 2, 4, 11191998, 2021, 112, 26, 61225, 1130, 61]
- Number of States: [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
- Dataset Features Included:
    - Motor Reliance: [GPM, Motor_RPM, Motor_Axial_Vibration, Motor_Lateral_Vibration, WOB]
    - Pulser Reliance: [GPM, Motor_RPM, Pulser_Axial_Vibration, Pulser_Lateral_Vibration, WOB]

- All Vibration: [Motor_Axial_Vibration, Motor_Lateral_Vibration, Pulser_Axial_Vibration, Pulser_Lateral_Vibration]
- Just Motor: [Motor_Axial_Vibration, Motor_Lateral_Vibration]
- Just Pulser: [Pulser_Axial_Vibration, Pulser_Lateral_Vibration]

It was important to include a sufficient number of random seeds to initialize the model such that a sufficient seed was found without too much bias to begin. The number of states was tested to see if there were models that would be better at modeling the data. Finally, the features from the original dataset that were included in the training and testing for a particular model was changed to see which combination would produce more accurate and less complicated models. The client suggested that vibration might be the most accurate predictor for the different states they were looking for. Therefore, the vibration sensed at the motor and the pulser was shifted around with a full set of other important features and amongst themselves in the five feature set variations.

Note that, for each implementation, the number of iterations was kept at a maximum of 200 and the covariance type was set to full. This is because any given feature set included dependent features and thus, a full covariance was needed.

After testing these models, it became clear that a final variation might be worth considering. In this variation, the only feature given to the model was WOB. This was because of the primary role WOB plays in actively drilling compared to the other activities that happen on a drilling rig that can be seen in the data recorded downhole. This final variation was considered due to the client's primary goal of identifying when drilling was occurring so their software could turn off and on as appropriate.

Evaluation

This project utilized the data from Flybar 1WB and Flybar 1WC to train the three methods. These methods were tested on the singular, but lengthy, run conducted on Flybar 2WC. The log likelihood of the models was used to represent which ones best fit the testing data, the AIC and BIC were used to determine the complexity of the models, and then a visual analysis of the four identified drilling parameters (flow, rotation, vibration, and WOB) was completed to assess the interpretability of the models' states.

Results

Over 700 models provided a wide variety of results. The top fifteen models in terms of log likelihood were significantly better than the other models created. Two feature sets outperformed the rest: WOB and Just Pulser, meaning data frames including only the axial and lateral vibrations measured at the pulser (Figure 2). Considering the domain knowledge used in choosing these parameters for basing the models on, it is not surprising that this was the case.

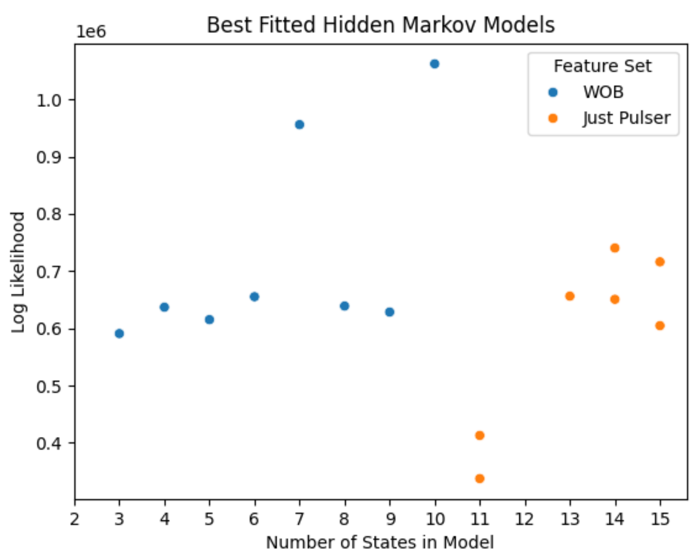

Figure 2. Top 15 HMMs with the highest log likelihoods

Furthermore, complexity is linearly scaled with the number of states in the model. The AIC for the top fifteen models identified above can be seen below (Figure 3). The graph for BIC looks almost identical.

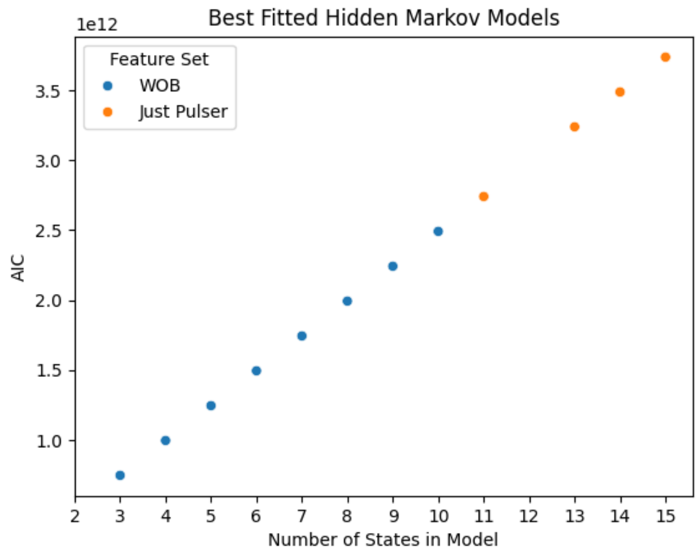

Figure 3. Complexity of the top 15 HMMs visualized with the AIC

With both of these parameters in mind, the most effective models tested include the WOB-based models with 7 and 10 states. Keep in mind that the WOB models were not tested for more than ten states, unlike the other feature sets due to the increasing difficulty in interpretability. Additionally, the WOB models were only tested with the random seed of 2021 in the interest of time.

Discussion

To better understand the 7- and 10-State WOB models, Primary Component Analysis (PCA) was conducted and the features for each state were compared. The features included in this analysis were GPM (flow), RPM (rotation), axial and lateral vibration, and WOB. Note that the vibration data came from the pulser rather than the motor because it is closer to the bit and was more effective for model building (Figure 2). For the purposes of this paper and to reduce redundancy, only the 7-State WOB Model will be represented here. This is due to the extreme similarities between the findings of each model and the easier visualization and interpretability of the 7-state model over the 10-state model.

The labeled testing data for the 7-State WOB Model was reduced to two dimensions using PCA and the loadings for each feature were found. PC1 explains approximately 32% of the variance in the data and PC2 explains about 24% of the variance in the data. Together, they explain about half of the data's variance. The layout of the clusters for each state and their PC values is seen in Figure 4. While it would have been more satisfactory to see a higher percentage of the variance explained by the first two components, an analysis with more dimensions seemed unnecessary since there were only five features to begin with.

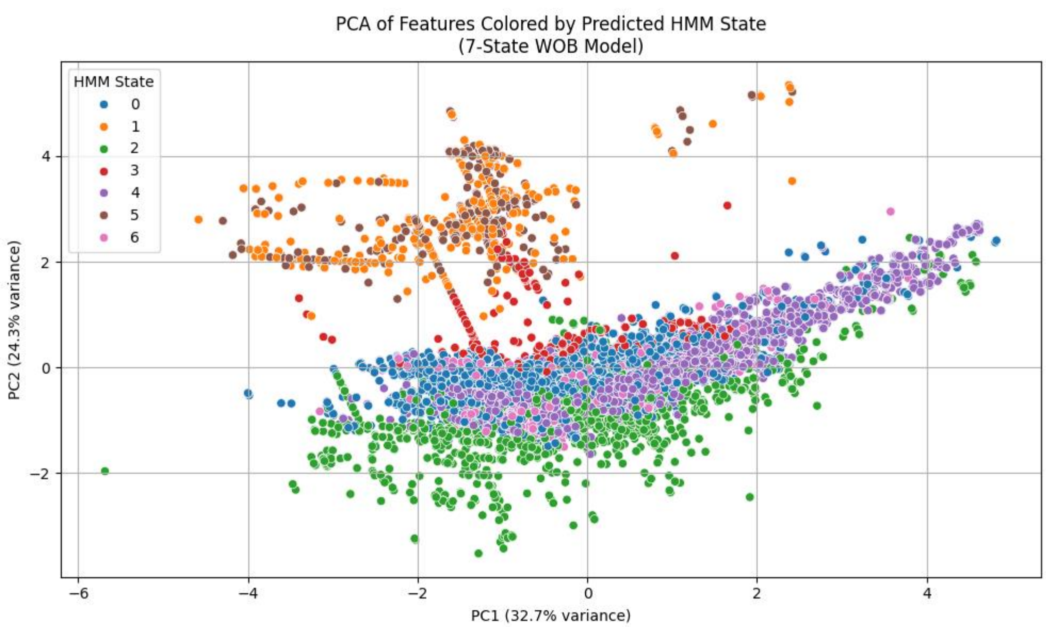

Figure 4. PCA of 7-State WOB Model (5% of data points represented)

The most influential features in defining these two principal components were the WOB, Axial Vibration, RPM (Figure 5). RPM is mentioned here over Lateral Vibration despite the latter's primary role in PC1 due to the higher aggregated effect of RPM. It also alleviates some of the concern about axial and lateral vibrations' interconnectedness in a closer manner than any other two features. It is no surprise that WOB was a key contributor to the separation of the data since the model was built on this feature. However, the roles of axial vibration and RPM require further investigation.

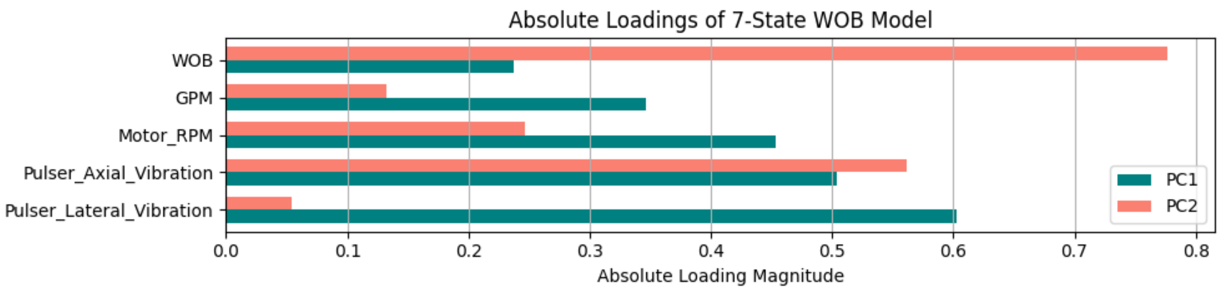

Figure 5. Loadings of features for primary components of 7-State WOB Model

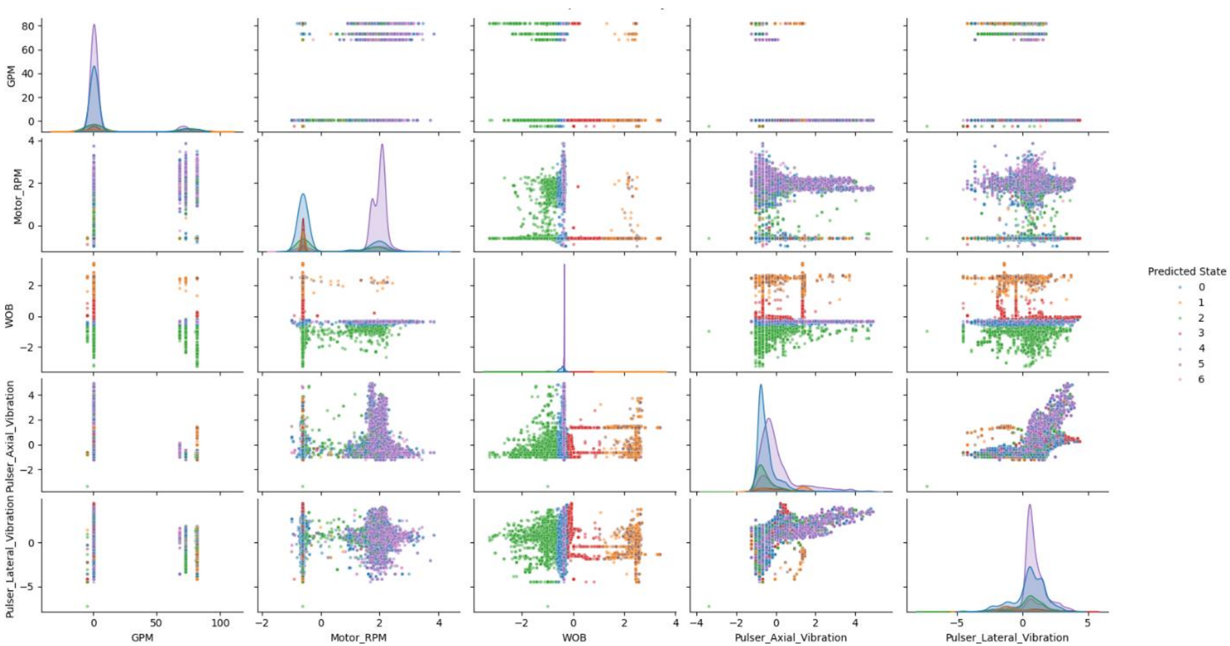

Figure 6. Feature comparison for 7-State WOB Model with color-coded states (5% of data points represented)

A visualization of the state clusters with a comparison of each of the features was created in order to better understand their roles in defining the states (Figure 6). The findings from this were rather inconclusive. At first glance, state-based patterns seem to emerge, especially on the WOB comparisons. However, a closer look reveals that the clusters are only based on WOB with no patterns relating to the second feature in the comparison.

Conclusion

The current implementation of the Hidden Markov Model was unsuccessful at identifying interpretable states that can be used for MWD software development. Because the states were not clustering around trends that would be expected for normal drilling activity changes and the subsequent parameter shifts, it is likely that the data was too sparse and the forward filling in the preprocessing obfuscated the true parameter patterns. Furthermore, a methodology with a higher level of interpretation is necessary to avoid the current result of highly suited models with no real-world application.

Future work on this project should investigate Gaussian Mixture Models (GMM) for clustering if no data with labels can be obtained. Furthermore, it would be worth looking into gathering accurately labeled data in order to circumvent any failures or difficulties in classification and turn directly towards creating supervised models, such as a Random Forest (RF) to directly implement into software that makes decisions on when to run different functions.

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